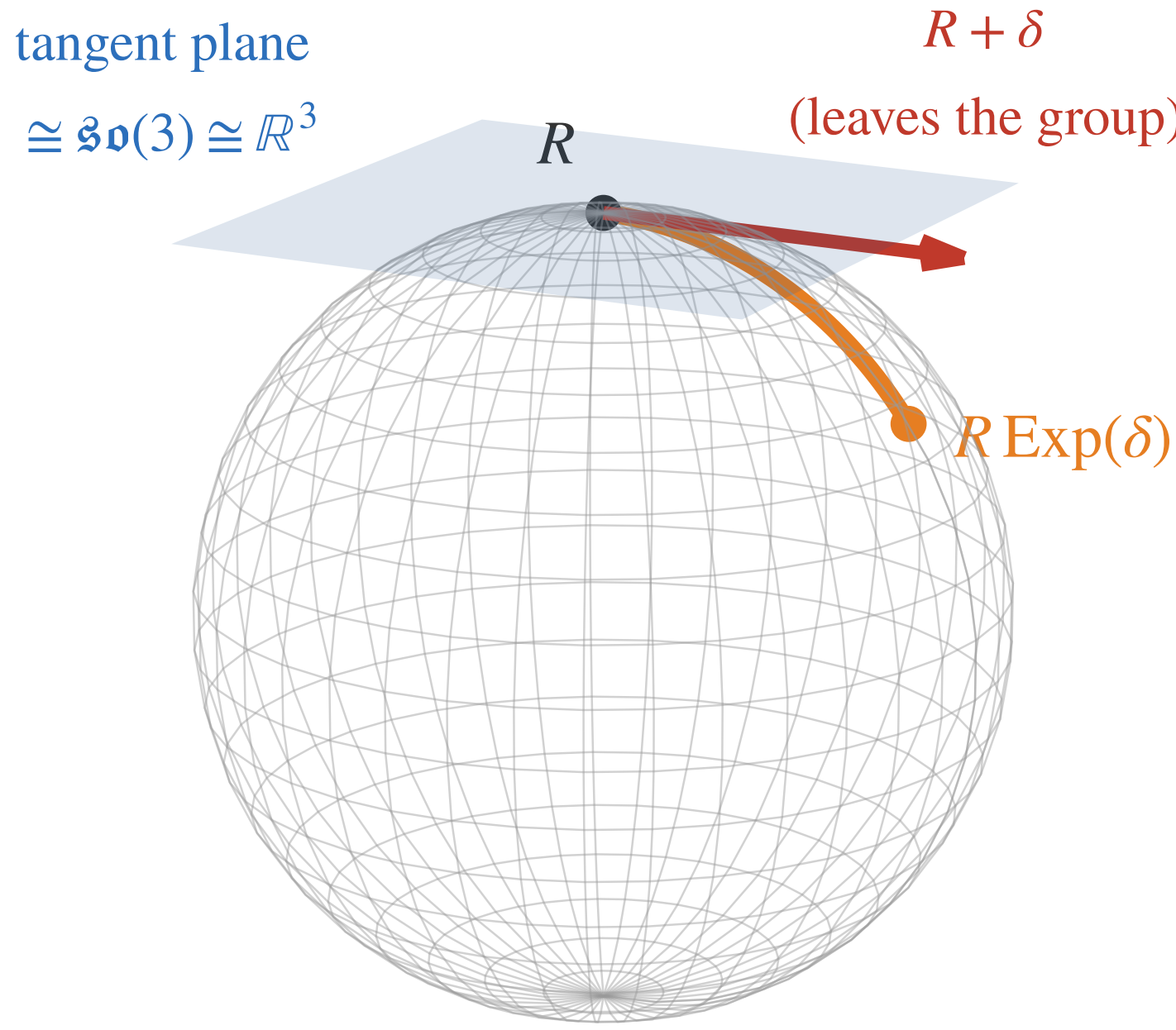
curved group, flat tangent space

(S^2 drawn as a stand-in for $SO(3)$)